An AI-Assisted Research Workflow for Retail Expansion

Turns fragmented public data into traceable site-screening recommendations and targeted follow-up research.

1. Situation

Retail expansion teams
shortlisting sites in a new city

Strategy consultants
desk phase of market-entry work

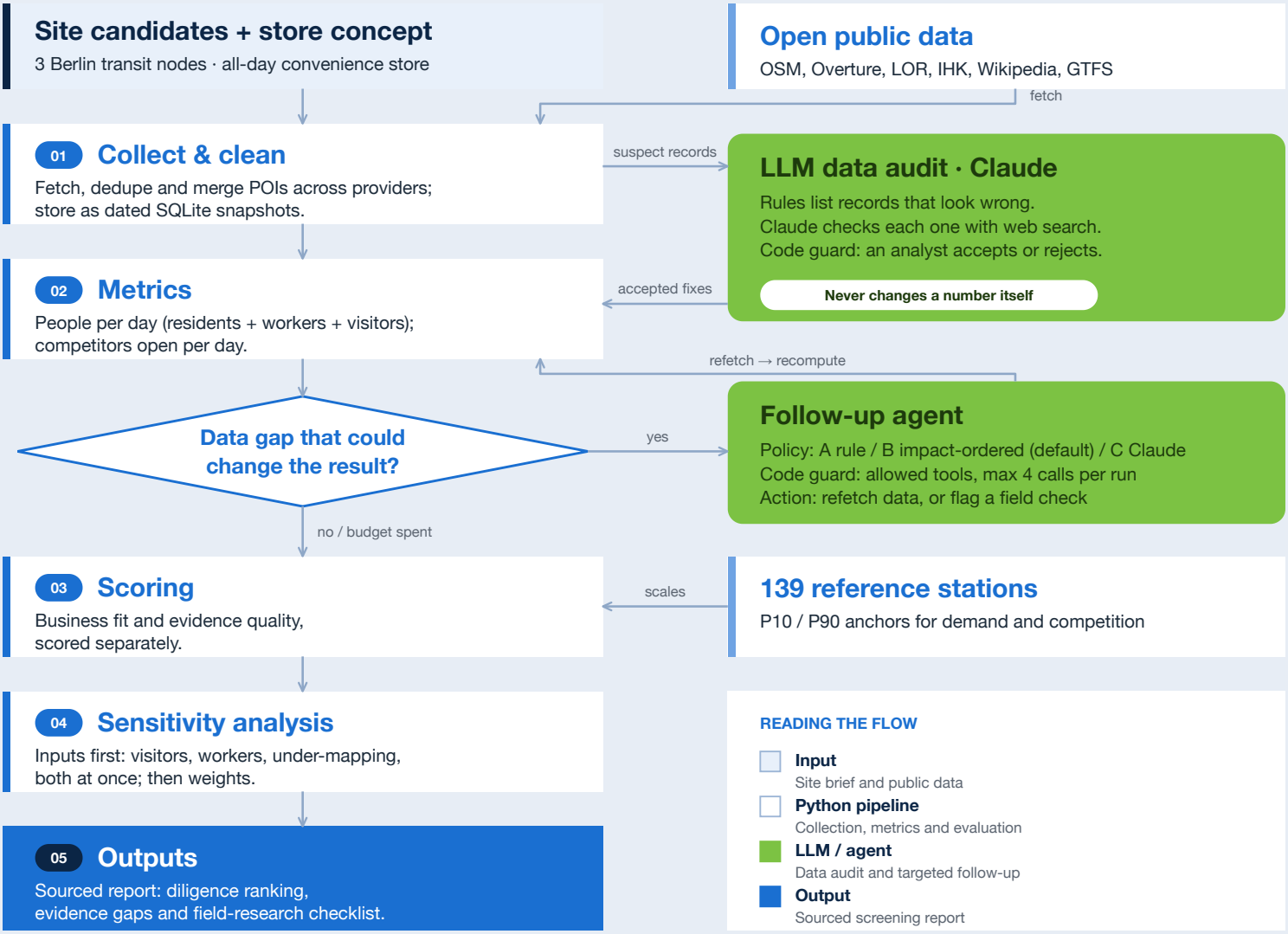
Location analysts
commercial and public-sector site studies

WHY DESK RESEARCH FALLS SHORT TODAY

- **Labour-intensive collection:** fragmented sources gathered and cross-checked by hand, site by site.
- **Misdirected follow-up:** verification effort goes to gaps that would not change the decision.
- **Inconsistent assessments:** different sources and criteria; results hard to compare or reproduce.
- **False precision:** a single ranking hides data gaps and untested assumptions.
- **Missing data read as zero:** no data for a site makes it look free of competition.

2. How it works

System architecture



Highlights

01
Deterministic scoring, bounded AI

Every score is computed deterministically. AI acts only within code-enforced budgets and guardrails; only reviewer-approved corrections reach the data.

02
City-wide benchmarking

Scales (P10 / P90) are calibrated on 139 Berlin S/U-Bahn stations, so each site is measured against the whole city, not just the other two.

03
Evaluation-led AI adoption

Both AI components passed pre-registered evaluations first. Rules matched the LLM on all 14 ordering decisions, so rules decide follow-ups and the LLM drafts field questions; the data audit met thresholds fixed in advance.

04
Production-grade engineering

241 automated tests, fully reproducible offline runs, and failure handling with rule fallback.

3. Case study

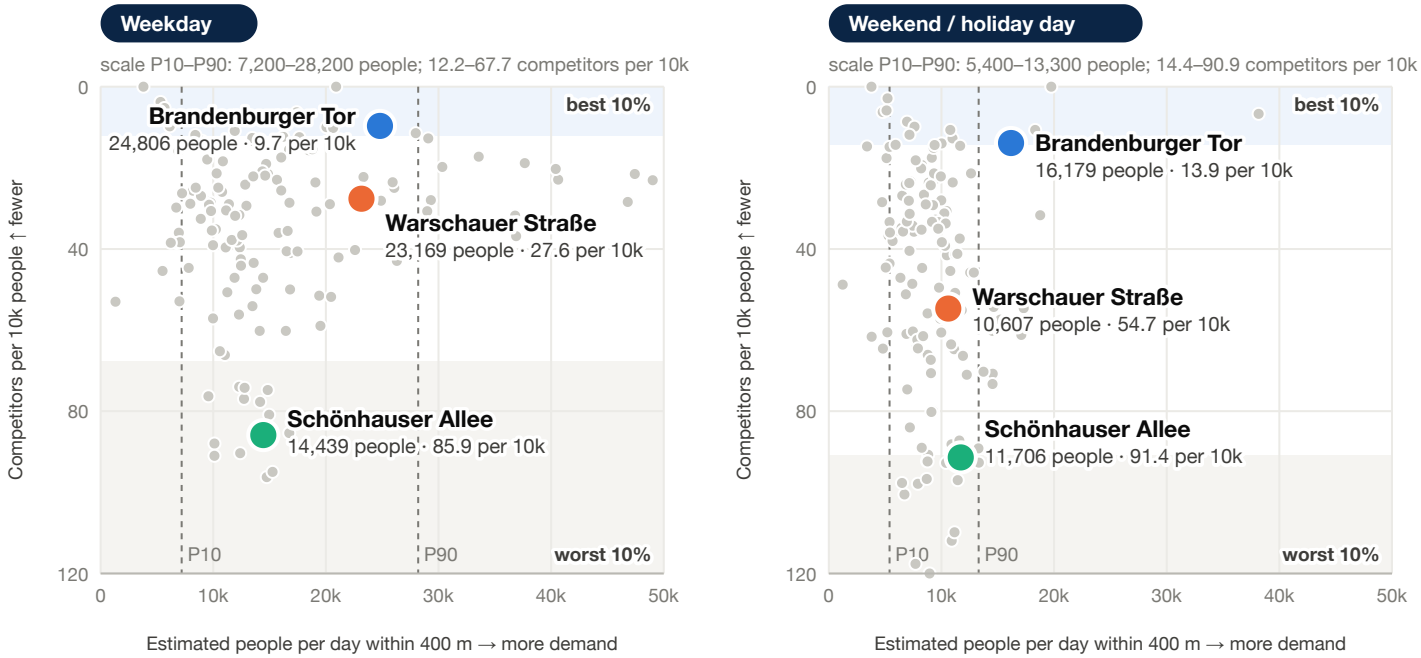
7-Eleven Due Diligence: 3 Berlin Transit Nodes

Hypothetical 7-Eleven convenience store: fresh food, coffee and daily goods, 50–100 m² at a station, trading all day.

- **Brandenburger Tor** · visitor-led centre
- **Warschauer Straße** · office + nightlife
- **Schönhauser Allee** · residential

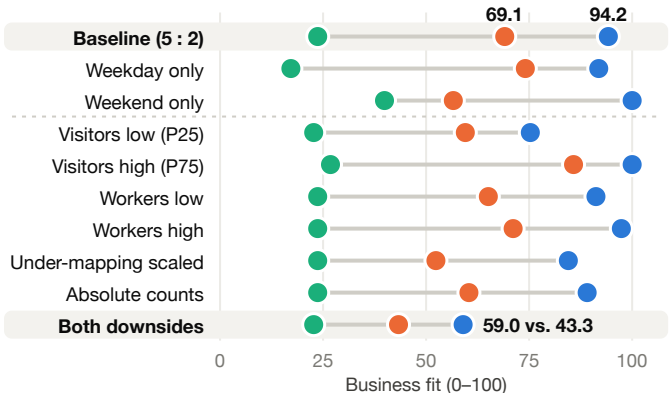
Measured against 139 Berlin stations, not just against each other

Grey dots: inner-Berlin S/U-Bahn stations; bands and dashed lines: each day type's P10 / P90. Brandenburger Tor is in the city's least contested tenth on both day types (9.7 / 13.9 competitors per 10,000 people); Schönhauser Allee in the most crowded tenth (85.9 / 91.4).



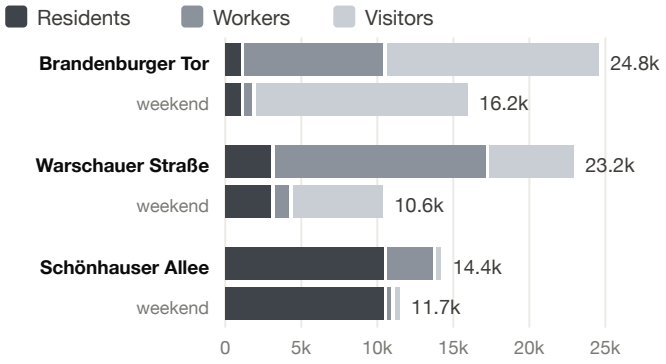
The ranking survives its least certain inputs

Brandenburger Tor is first in all 10 stress tests, including both downsides at once (59.0 vs. 43.3).



Each node draws a different crowd

People per day within 400 m (upper bar weekday, lower weekend). Visitors are 57% of Brandenburger Tor's weekday demand, the figure to verify first.



4. Manual vs. tool

● Manual / analyst ● Tool

×1,900 · TIME

32 min → 1 s

Competitor inventory for one node.

CITY SCALE

74 h → 1.5 s

Competitor inventories for all 139 reference stations.

LABOUR

9 → 2

Analyst touchpoints per node: 9 manual steps become 2 reviews (AI flags, field visits).

COST

€0

Marginal cost of a rerun; optional AI audit $0.25–1.25 per site.

Evidence: one manual benchmark (one analyst, one node, not blind). The 139-station manual time is extrapolated at 32 min each; tool times are measured (139 stations from cached data).

Tech stack Python · SQL / SQLite · API integration · Anthropic Python SDK

Engineering & evaluation AI-assisted development with Claude Code · LLM tool calling · LLM evaluation · Automated testing · GitHub Actions CI